

Tianyi Xu

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Education

University of Wisconsin-Madison

Sept 2021 – Dec 2025

BS in Computer Science, Data Science, Mathematics

- GPA: 3.918/4.000
- **Coursework:** Foundation Models, Deep Learning in Computer Vision, Learning Based Image Synthesis and Manipulation, Data-Centric ML, Real Analysis, Probability Theory, Theoretical Data Science

Research Experiences

Research Assistant

Madison, WI

Advisor: Junjie Hu

July 2025 – Present

- Developed **tone-aware, speaker-invariant** speech representations for **low-resource tonal languages** by adapting multilingual self-supervised speech encoders (wav2vec2/ XLS-R) using fairseq.
- Built a **multitask training** pipeline with contrastive learning and ASR CTC auxiliary loss to reduce speaker bias while preserving tonal distinctions, regularized with **knowledge distillation**.
- Benchmarked evaluation via **cross-speaker retrieval**, achieving **0.95-0.99** Top-1/Top-5 accuracy consistently while maintaining good tone distinction. Manuscript in preparation for ACL 2026.

Research Intern, Epic Lab (SJTU)

Remote

Advisor: Linfeng Zhang, Mentor: Shaobo Wang

Aug 2025 – Present

- Built a **multi-GPU** GPT-2 pretraining pipeline (distributed training) and implemented Llama3/Qwen3 pretraining with **Megatron** framework.
- Developed **online, validation-guided data selection** and **domain-mix adaptation** modules, and integrated them with optimizers such as **Adam** and **Muon** for stable long-run training.
- Ran ablation studies on data mixes/selection cadences; evaluated on **MMLU**, **HellaSwag**, **OpenBookQA**, etc; improving accuracy for **+2–3 points** on average. Manuscript in preparation for ICML26.

Research Assistant

Madison, WI

Advisor: Claudia Solís-Lemus

Jan 2023 – Present

- Analyzed **large-scale, high-dimensional microbiome datasets** using **visualization, dimensionality reduction, regression/statistical modeling**, and **network analysis** in **R** to identify ecological and microbial association patterns.
- Developed a **label-efficient bioacoustic detection** pipeline for biodiversity monitoring: pretrained an **AudioMAE** encoder on unlabeled rainforest soundscapes and integrated it into a detector with **self-training**, improving **~0.3 mAP** over supervised baselines under out-of-distribution site/time shifts.

Independent Research Study

Madison, WI

Advisor: Pedro Morgado

Jan 2025 – May 2025

- Built a **latent generative pipeline** for vision-guided audio separation (VAE latent space) that cuts compute/memory while preserving quality.
- Implemented and compared **latent diffusion** and **rectified flow matching** with visual conditioning; introduced **per-channel z-norm + clipping** that stabilized training.
- Built a visual-guided audio diffusion pipeline with U-Net conditioning; evaluated with FAD, LPAPS, CLAP, and ImageBind on AVE/Music datasets.

Publications

OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration ([Blog](#)) [🔗](#)

Under Review

Shaobo Wang*, Xuan Ouyang*, **Tianyi Xu***, Yuzheng Hu, Jialin Liu, Guo Chen, Tianyu Zhang, Yuzheng Hu, Junhao Zheng, Yubo Ma, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang

Submitting to ICML 2026

SITA: Learning Speaker-Invariant and Tone-Aware Speech Representations for Low-Resource Tonal Languages ([arXiv](#)) [🔗](#)

Under Review

Tianyi Xu*, Xuan Ouyang*, Binwei Yao, Shoua Xiong, Sara Misurelli, Maichou Lor, Junjie Hu

Submitted to ACL 2026.

Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring

Under review

Tianyi Xu, Claudia Solís-Lemus, Daniel Pimentel-Alarcon, Zuzana Burivalova

Under review at CV4EO, WACV 2026

In preparation for Journal Submission.

Combined effects of methyl bromide and soil amendments on soil bacterial and fungal communities in turfgrass ([Preprint](#)) [🔗](#)

Under review

Tianyi Xu*, Salma Mukhtar*, Evan Gorstein, Claudia Solís-Lemus, Ming Yi Chou, Paul Koch

Submitted to Soil Use and Management.

Stem rot affects the structure of rhizosphere microbiome in berseem clover (*Trifolium alexandrinum*) (Published in Rhizosphere)

Apr 2025

Salma Mukhtar, Zain Ahmad, Noor Khan, **Tianyi Xu**, Dalaq Aiysha

[10.1016/j.rhisph.2025.101073](https://doi.org/10.1016/j.rhisph.2025.101073) [🔗](#)

BiomedBank: A Large-scale, Multimodal Data Ecosystem for Advancing Biomedical AI

In preparation

TBD

Manuscript in preparation.

Talks / Presentations

Friday at 4 Seminar – Invited Talk

Dec 2025

“Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring”

Department of Plant Pathology, University of Wisconsin–Madison

Undergraduate Research Symposium – Poster Presentation

Apr 2025

27th Annual Undergraduate Symposium, University of Wisconsin–Madison

“Biodiversity Detection Using Acoustic Signals with Deep Learning”

Presented as part of the requirement of Holstrom Environmental Fellowship on the progress of the project.

[Abstract Link \(Page 164\)](#) [🔗](#)

Undergraduate Research Symposium – Poster Presentation

Apr 2024

26th Annual Undergraduate Symposium, University of Wisconsin–Madison

“Biodiversity Detection Using Acoustic Signals with Deep Learning”

[Abstract Link \(Page 139\)](#) [🔗](#)

Awards

Phi Beta Kappa (PBK), Inducted

2025

Holstrom Environmental Research Fellowship

2024

Awarded \$3,000 + \$1,000 (advisor) for research on acoustic event detection in tropical rainforests using deep learning.

UW–Madison Summer Scholarship

2024

Awarded \$1,000 for Summer 2024 studies in recognition of outstanding academic achievement.

Dean's List (Six Semesters)

2021–2024

Recognized for maintaining a GPA above 3.85 in Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2024, and Fall 2024.

Other Experiences & Projects

Project Assistant / Intern Remote

Nov 2024 – May 2025

- Contributed to *BiomedBank*, a large-scale multimodal biomedical dataset project, collaborated with Prof. Lichao Sun(Lehigh) and Prof. Chaowei Xiao(UW-Madison/JHU).
- Conducted literature review on data quality and built preprocessing/deduplication pipeline for biomedical NLP. Benchmarked biomedical QA and Clinical Trial Dataset with VLLM deployed medical/general Large Language Models (e.g Med-LlaMa3, GPT-OSS, Qwen3).

Project Assistant Wisconsin Alzheimer's Disease Research Center

Sept 2023 – Aug 2024

- Developed diagnosis prediction pipeline with SMOTE, Random Forest, and XGBoost.
- Explored CNN and phylogenetic embeddings for feature representation; used VAE for data synthesis.

Peer Mentor UW-Madison Department of Computer Science

Sept 2022 – May 2024

- Mentored 400+ students in CS 220, 320, and 540 on Python, AI, and data science topics.
- Hosted 400+ office hours covering OOP, visualization, and ML/DL fundamentals.

Developer UW-Madison DoIT Helpdesk

Oct 2021 – Aug 2022

- Improved UI/UX with dark mode enhancements and user preference handling.
- Built and documented APIs integrated with MySQL for efficient data access.
- Created staff filtering system in React.js to support searches across 1000+ employees.

Selected Graduate Course Projects/Papers University of Wisconsin-Madison

2023-2024

- *ABTUNE – Alpha Batch Size Tuning (Graduate Foundation Model Course, Full Mark)*: Benchmarked two-phase training and adaptive batch size strategies on noisy vs. clean data across image and text classification.
- *Learning Smartly – Data-Centric Methods for LMs (Graduate-Level Data-Centric ML Course, Full Mark)*: Developed a hybrid framework combining LLM-generated label functions and active learning to improve weak supervision in text classification tasks.
- *Image Style Transfer with Diffusion Models (STAT 453 Deep Learning, A)*: Supervised fine-tuned (SFT) Stable Diffusion using curated paired datasets (CycleGAN, AdaIN, GPT-4 prompts) for semantic-preserving, text-guided multi-style transfer.
- *ImageEditNet – SFT on InstructPix2Pix (Graduate Learning-Based CV Course, A)*: Improved real-image text-guided editing via SFT on MagicBrush and GAN-generated datasets, boosting CLIP similarity and facial editing quality.
- *ASL Fingerspelling Recognition (CS 539 Neural Network, Full Mark)*: Fine-tuned Faster R-CNN (ResNet50, MobileNetV2) for real-time ASL letter detection, achieving 0.913 IoU and 0.7 accuracy; deployed webcam demo for live inference.

Skills

Programming Python, R, C/C++, Java, SQL, JavaScript, HTML/CSS

ML/DL: PyTorch, TensorFlow, Hugging Face (Transformers, Diffusers), PyTorch Geometric, Scikit-Learn

LLM Training / Systems: Megatron(-LM), distributed training (multi-GPU), CUDA

Data / Infra: Pandas, Spark, HDFS, BigQuery, Cassandra, Kafka

Tools: Git, Linux, Docker, GCP, MySQL, SQLite, Flask, Selenium, Streamlit, Snowflake, Elasticsearch, Airbyte, Weights & Biases

Languages: Chinese (Mandarin, native), English (fluent), Japanese (basic)